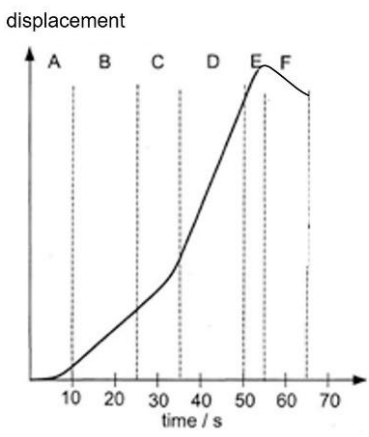


2019 Prelim Exams Paper 3 Answers

1	(a)	(i)	In Section E, the object undergoes a <u>constant deceleration of 7 m s^{-2}</u> until velocity becomes zero at $t = 55 \text{ s}$, after which the particle <u>reverses its direction</u> as the velocity changes from positive to negative. In Section F, the particle travels at a <u>constant velocity of 5 m s^{-1}</u> in the direction <u>towards the starting point</u>.	B1 B1
		(ii)	 Sections B, D and F - Linear graphs Sections A and C – increasing distance at increasing rate Section E – increasing distance at decreasing rate, then decreasing distance (allow ecf from 1(a)(i) for Section E)	B1 B1 B1
	(b)	(i)	Consider the projectile motion from B to C: $S_y = u_{By}t + \frac{1}{2}at^2$ $1 = u_{By}t + \frac{1}{2}(9.81)t^2$ $1 = u_B \sin 30^\circ t + \frac{1}{2}(9.81)t^2 \quad \text{----Eqn (1)}$ $S_x = u_{Bx}t$ $1 = u_B \cos 30^\circ t$ Rearranging, $t = \frac{1}{\cos 30^\circ u_B}$ Substitute into equation (1) $1 = u_B (0.5) \frac{1}{\cos 30^\circ u_B} + \frac{1}{2}(9.81) \left(\frac{1}{\cos 30^\circ u_B} \right)^2$ $1 = 0.57735 + \frac{6.5399}{U_B^2}$ Solving, $U_B = 3.9 \text{ m s}^{-1}$	M1 M2 M3
		(ii)	Lost in GPE = gain in KE $mgh = \frac{1}{2}m(U_B)^2$ $h = 0.775 \text{ m}$	M1 A1